

MONKEY BANANA PROBLEM

AIM

To write a Prolog program to implement the Monkey Banana Problem using logical rules to determine how the monkey can reach and obtain the bananas.

ALGORITHM

1. Start.
2. Define the initial position of the monkey, box and bananas.
3. Define possible actions such as moving, pushing the box and climbing.
4. Check whether the monkey is under the bananas.
5. If not, move or push the box.
6. Climb the box.
7. Grab the bananas.
8. Display the sequence of actions.
9. Stop.

SOURCE CODE

```
% Monkey Banana Problem
```

```
move(monkey, door).
```

```
move(monkey, window).
```

```
push_box :-
```

```
    write('Monkey pushes the box under the bananas. '), nl.
```

```
climb_box :-
```

```
    write('Monkey climbs onto the box. '), nl.
```

```
grab_banana :-
```

```
    write('Monkey grabs the bananas. '), nl.
```

get_banana :-

move(monkey, door),

push_box,

climb_box,

grab_banana.

OUTPUT

```
% c:/Users/Abinaya Sri J/OneDrive/Desktop/Prolog exercise
?- get_banana.
Monkey pushes the box under the bananas.
Monkey climbs onto the box.
Monkey grabs the bananas.
true.
?-
```

RESULT

The Prolog program successfully demonstrates the sequence of actions required for the monkey to obtain the bananas.